

# Classic Error Practice & Teacher Mnemonics Flashcards

Student Name: \_\_\_\_\_

Cycle: Semester 1, Weeks 3-4 Daily Goal: Complete 2 cards per day, master the core concepts

## Card 01

Divisibility Modulo · William's Focus

💡 **Mnemonic: Split 72 into 8 and 9! Last 3 digits determine Y, sum of all digits determines X. Complete both steps!**

### [Parallel Practice Challenge]

The six-digit number  $\overline{73A64B}$  is divisible by both 8 and 9. Find  $A$  and  $B$ .

① Last 3 digits:  $\overline{64B} \div 8$  has no remainder, so  $B =$  \_\_\_\_\_;

② Sum of digits:  $7 + 3 + A + 6 + 4 + B$  is a multiple of 9, so  $A =$  \_\_\_\_\_.

[Key]:  $B = 0$  or  $8$ . If  $B = 0$ ,  $A = 7$  (737640); if  $B = 8$ ,  $A = 8$  (738648).

## Card 02

Decimal Addition & Division · William

💡 **Mnemonic: Ten tenths make one whole!  $0.9 + 0.1 = 1.0$ . In division, shift the decimal point equally:  $5.4 \div 0.6 = 54 \div 6 = 9$ !**

### [Mental Speed Drill]

(1)  $0.8 + 0.2 =$  \_\_\_\_\_

(2)  $4.9 + 0.1 =$  \_\_\_\_\_

(3)  $7.2 \div 0.8 =$  \_\_\_\_\_ (⚠️ NOT 0.9!)

(4)  $40 \div 0.5 =$  \_\_\_\_\_

[Key]: (1) 1.0; (2) 5.0; (3) 9; (4) 80.

## Card 03

Powers vs Products · Both Students

💡 **Mnemonic: Base multiplies itself!  $2^3$  means three 2s multiplied:  $2 \times 2 \times 2 = 8$ , NEVER  $2 \times 3 = 6$ !**

### [Write Out Full Expanded Form]

(1)  $2^3 =$  \_\_\_\_\_  $=$  \_\_\_\_\_

(2)  $3^4 =$  \_\_\_\_\_  $=$  \_\_\_\_\_

(3)  $10^3 \times 10^4 = 10^{+} = 10^{+}$

[Key]: (1)  $2 \times 2 \times 2 = 8$ ; (2)  $3 \times 3 \times 3 \times 3 = 81$ ; (3)  $10^7$ .

## Card 04

Square Numbers · Sophia's Focus

💡 **Mnemonic: Twin identical factors make a square!  $2 \times 3 = 6$  is a rectangle, so 6 is NEVER a square!**

### [Identification & Geometry]

(1) Circle the square numbers:

12, 16, 18, 25, 27, 36, 45, 49

(2) A square has an area of  $49 \text{ cm}^2$ . What is its side length?

Answer: Side length = \_\_\_\_\_ cm (Do NOT write  $49 \div 2$ ).

[Key]: (1) Square numbers: 16, 25, 36, 49; (2) Side length = 7 cm ( $7 \times 7 = 49$ ).

**Card 05****Fraction Expansion · Sophia's Focus**

💡 Mnemonic: What you do to the bottom, you **MUST** do to the top! Never leave the numerator unscaled!

**[Find and Fix the Error]**

Mistake:  $\frac{2}{3} + \frac{1}{4} = \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}$

Correct solution:

$$\frac{2}{3} + \frac{1}{4} = \frac{\quad}{12} + \frac{\quad}{12} = \underline{\hspace{2cm}}$$

[Key]: Expression =  $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$ .

**Card 06****Smart Fraction Comparison**

💡 Mnemonic: Equal numerators? Smaller denominator is bigger! Close to 1? Smaller residual distance is bigger!

**[Fill with > or <, state rule]**

(1)  $\frac{9}{16}$  \_\_\_\_\_  $\frac{9}{20}$  (Same numerators)

(2)  $\frac{19}{20}$  \_\_\_\_\_  $\frac{29}{30}$  (Compare distance to 1)

[Key]: (1) >; (2) < ( $1 - \frac{19}{20} = \frac{1}{20} > 1 - \frac{29}{30} = \frac{1}{30}$ ).